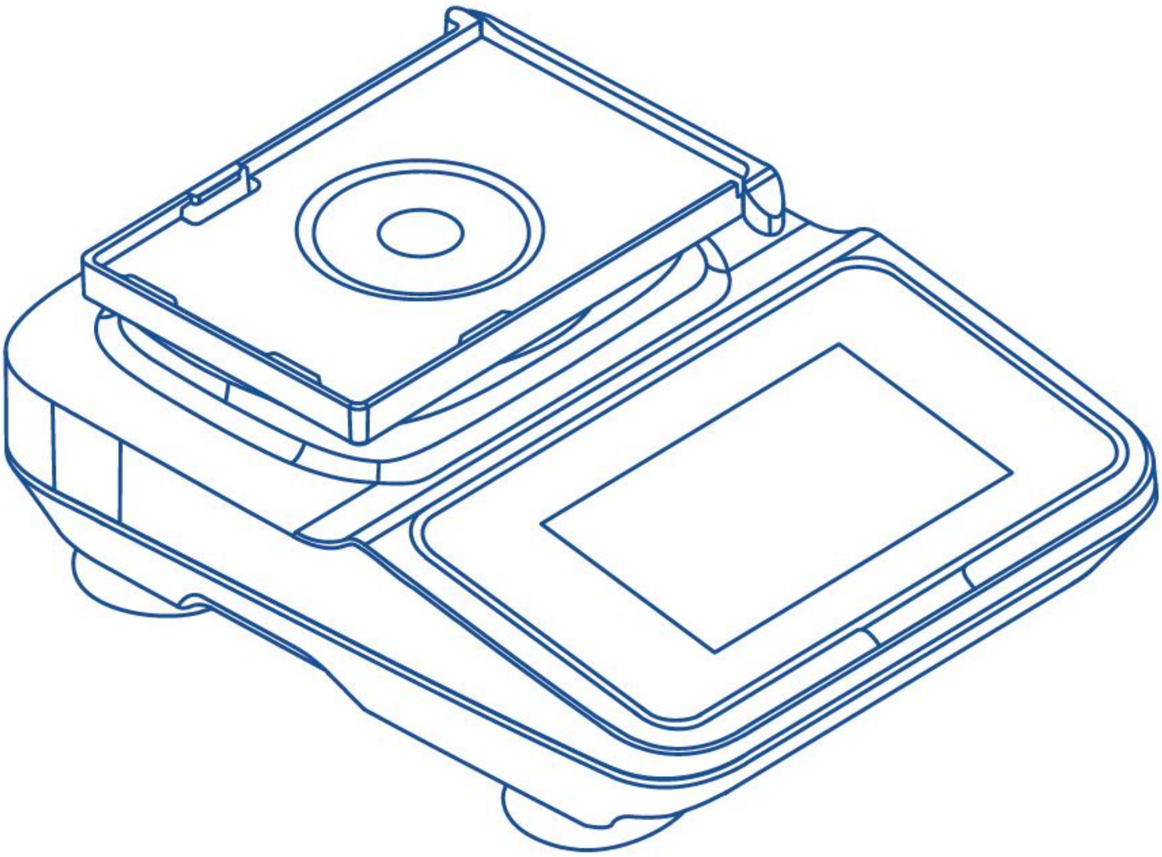
VORTEX MIXER
智能涡旋仪

OPERATION MANUAL
操作手册



QUALITY
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Contents

3. Safety Instructions

3.1 Operating Environment

3.1.1 Indoor use only.

3.1.2 Altitude ≤ 2000m.

3.1.3 Applicable ambient temperature range: 5-40°C.

3.1.4 Applicable relative humidity range: ≤80%.

3.1.5 Power supply applicable to select product models.

3.1.6 Adequate ventilation equipment must be installed indoors.

3.1.7 Ensure there is no vibration or airflow in the surrounding area that affects performance.

3.1.8 Ensure the surrounding air contains no conductive dust, explosive gases, or corrosive gases.

3.2 Safety Tips

3.2.1 Please read this manual carefully before using the device for the first time.

3.2.2 The metal bath may only be operated by trained and authorized personnel.

3.2.3 Equipment maintenance must only be performed by our company or our authorized agents.

3.2.4 The following materials are strictly prohibited in the instrument:

— Flammable and explosive materials;

— Strongly reactive chemical materials;

— Toxic, radioactive substances, or pathogenic microorganisms.

3.2.5 Only qualified maintenance personnel using appropriate tools may perform repair operations on the metal bath system.

3.2.6 If the instrument operator encounters situations not covered in this manual, please contact our company or our authorized agents for guidance on the correct handling procedure.

3.2.7 Use accessories provided by our company whenever possible (modules, lids, heated lids, etc.). If the user uses other accessories, our company will not be liable for any unintended consequences.

3.2.8 The instrument must be inspected and maintained at specified intervals.

3.3 Precautions

3.3.1 Do not plug/unplug the power cord or toggle the power switch with wet hands!

3.3.2 Do not plug/unplug the power cord while the instrument is powered on!

3.3.3 Do not maintain or clean the instrument while it is powered on!

3.3.4 Do not touch the module with bare hands or remove samples directly when the module temperature is high. Beware of high-temperature burns and low-temperature frostbite. It is recommended to use the "Return to Room Temperature" function before removing samples!

3.3.5 Do not install the instrument on uneven, swaying, or vibrating work surfaces!

3.3.6 Do not operate the instrument without a module installed. When the module is not properly installed, the module model displays "Module Not Detected," and the Run/Pause/Stop buttons are all disabled!

3.3.7 The lower heating plate temperature must not exceed 108°C (physical limit; output may stop if exceeded)!

3.3.8 When the module temperature is below 20°C, the hot lid function is disabled. Please keep samples insulated.

4. Device Introduction

4.1 Product Overview

The B5 PRO Intelligent Metal Bath features a 7-inch DGUS serial touch screen control interface and supports both heating and cooling modes for precise temperature control, with a temperature control range of 23~100°C and a temperature setting accuracy of 0.1°C. The instrument uses a PT1000 platinum RTD temperature sensor (4 channels) and an AD7124 24-bit high-precision ADC, combined with a dual-loop cascaded PID control algorithm (incremental + positional), ensuring high-precision temperature control even during long-term operation. The unit supports both manual and program operating modes and can store up to 50 programs (each with up to 10 steps). The operation curve page displays temperature trends in real time, and the instrument also provides running logs, alarm records, and USB/SD data export functions. It is compatible with centrifuge tube modules and multi-well plate modules of various specifications; modules can be replaced easily without tools. A standard module lid is included, and an optional heated lid is available to prevent low-temperature condensation.

4.2 Model and Technical Specifications

4.2.1 Product Model: B5 PRO

4.2.2 Product Name: B5 PRO Intelligent Metal Bath

4.2.3 Temperature Range (°C): 23-100

4.2.4 Temperature Setting Accuracy (°C): 0.1

4.2.5 Control Method: Dual-loop cascaded PID (incremental + positional)

4.2.6 Operation Method: Manual/Program

4.2.7 Timer Function: 99h 59min 59s

4.2.8 Display: 7-inch DGUS serial touch screen

4.2.9 Main Control Chip: STM32F407VETx (168MHz Cortex-M4F)

4.2.10 Operating System: FreeRTOS 10.x real-time OS

4.2.11 Temperature Sensor: PT1000 platinum RTD (4 channels)

4.2.12 ADC Accuracy: AD7124 24-bit high-precision ADC

4.2.13 Storage: FRAM FM24V05 64KB (non-volatile)

4.2.14 Communication Protocol: Modbus, DGUS serial display protocol

4.2.15 Heating Performance: Three heating modes (Full Power: >5°C/min; High: 5°C/min; Medium: 2°C/min); heating time ≤15 min (room temp 25°C → 100°C)

4.2.16 Cooling Performance: Three cooling modes (Full Power: ≥2°C/min; High: ≥1°C/min; Medium: ≥0.5°C/min); cooling time ≤15 min (room temp 100C → 23°C)

4.2.17 Programs: Up to 50 programs, each with up to 10 steps

4.2.18 Log Capacity: Up to 20 entries (4 pages) of run logs + up to 20 entries (4 pages) of alarm records

4.2.19 Temperature Curve: Up to 12,800 data points

4.2.20 Compatible Modules: 0.6 / 1.5 / 2 / 5 / 15 / 50 mL centrifuge tubes; multi-well plate modules (cell culture plates, ELISA plates, full-skirted microplates)

4.2.21 Compatible Lids: Standard module lid included (precision temperature control); optional heated lid (prevents low-temperature condensation, 1.5/2mL)

4.2.22 Software Scale: Approximately 21,000 lines of C code

4.3 Device Components

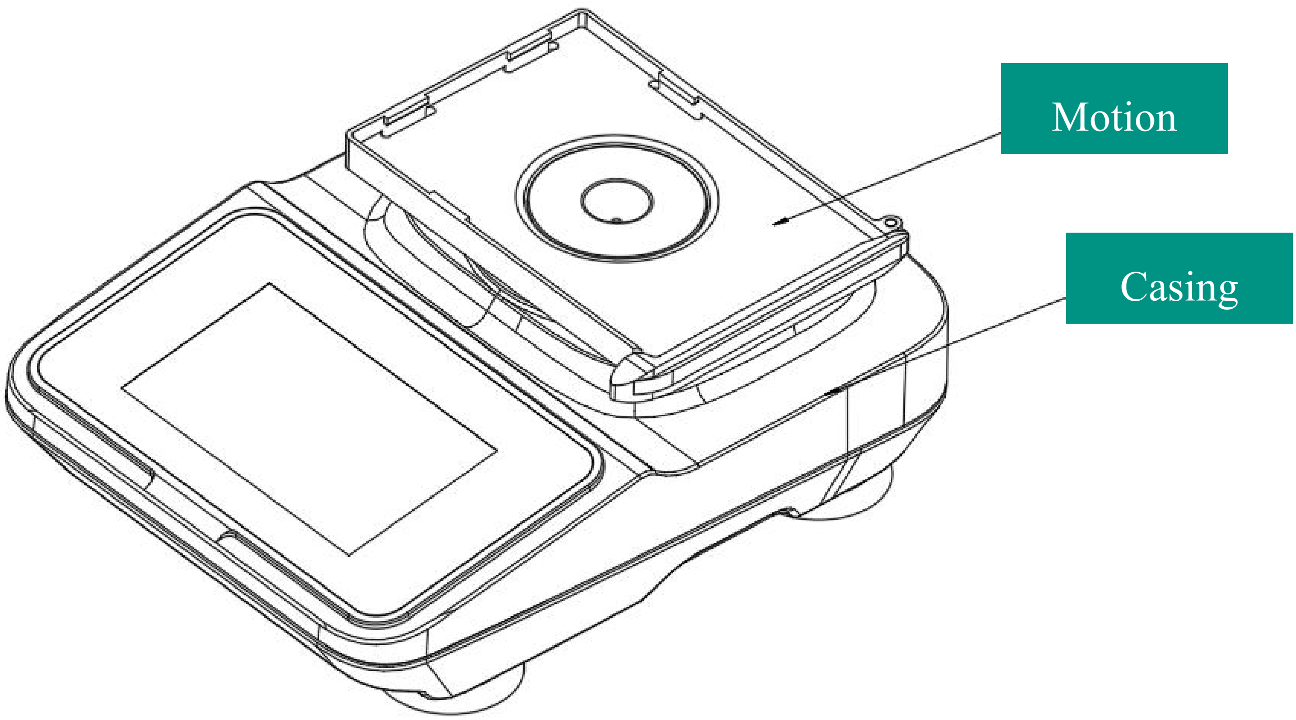
The B5 PRO Intelligent Metal Bath mainly consists of the following parts:

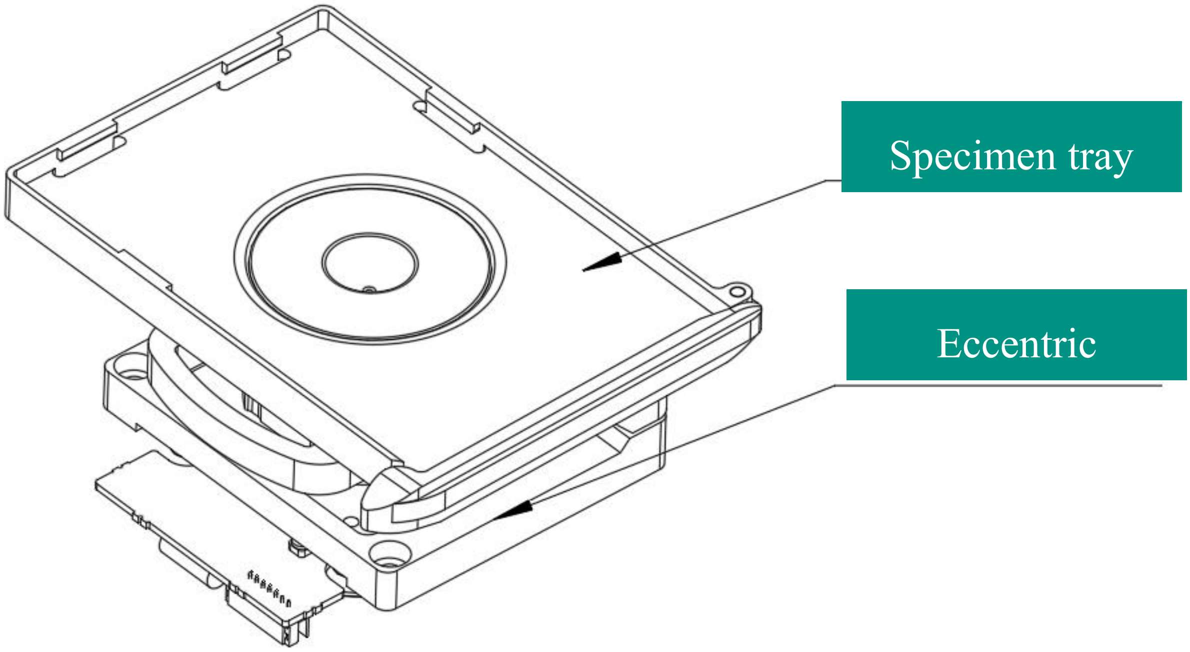
(1) Main unit: Built-in heating/cooling system and control board, equipped with a 7-inch touch screen, providing USB and SD card interfaces;

(2) Sample modules: Centrifuge tube modules and multi-well plate modules of various specifications, automatically identified via hardware ID pins (up to 16 module types supported), tool-free replacement;

(3) Module lid: Standard module lid included (precision temperature control); optional heated lid (prevents low-temperature condensation, compatible with 1.5/2mL modules);

(4) Power cord.



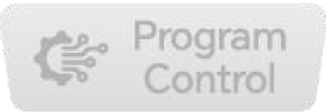
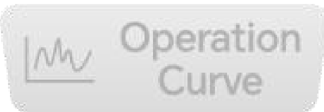
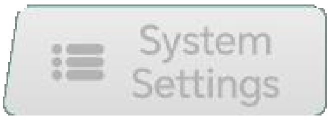


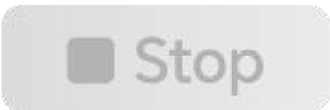
4.4 Function Introduction

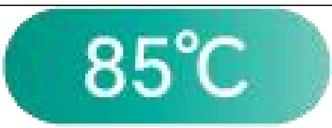
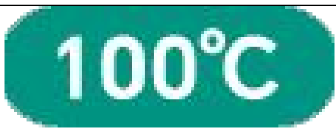
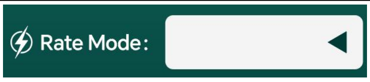
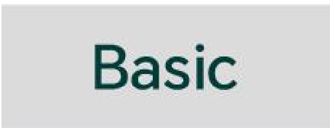
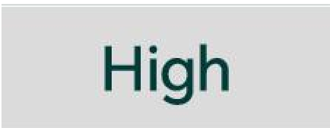
4.4.1 Icon Definitions

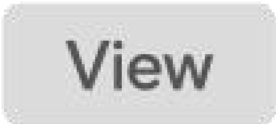
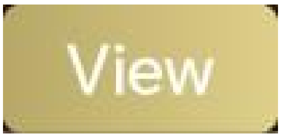
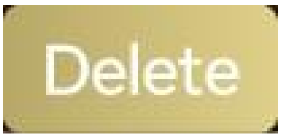
The main icons on the touch screen interface and their meanings are shown in the table below:

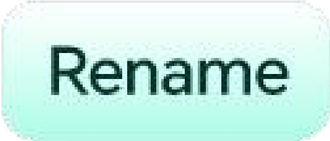
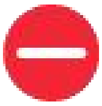
Icon	Icon Meaning
	Manual Setup (Highlighted)
	Program Control (Highlighted)
	Program Edit (Highlighted)
	Operation Curve (Highlighted)
	System Settings (Highlighted)

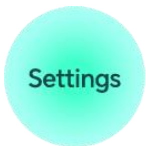
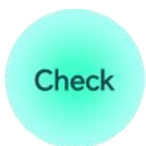
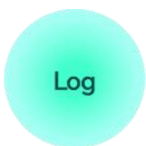
Icon	Icon Meaning
	Manual Setup (Normal)
	Program Control (Normal)
	Program Edit (Normal)
	Operation Curve (Normal)
	System Settings (Normal)
	Manual Setup (Disabled)
	Program Control (Disabled)
	Program Edit (Disabled)
	Operation Curve (Disabled)
	System Settings (Disabled)
	Warning message - Please read before operation
	Start (Highlighted)

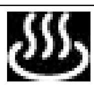
Icon	Icon Meaning
	Pause (Highlighted)
	Stop (Highlighted)
	Start (Normal)
	Pause (Normal)
	Stop (Normal)
	Stop (Disabled)
	Pause (Disabled)
	Stop (Disabled)
	Common Temp: 37°C (Normal)
	Common Temp: 65°C (Normal)
	Common Temp: 85°C (Normal)
	Common Temp: 100°C (Normal)

Icon	Icon Meaning
	Restore Room Temp (Normal)
	Common Temp: 37°C (Highlighted)
	Common Temp: 65°C (Highlighted)
	Common Temp: 85°C (Highlighted)
	Common Temp: 100°C (Highlighted)
	Restore Room Temp (Highlighted)
	Rate Mode: Basic, High, Extreme
	Module Model
	Basic (Normal)
	High (Normal)
	Extreme (Normal)
	Basic (Highlighted)

Icon	Icon Meaning
	High (Highlighted)
	Extreme (Highlighted)
	Confirm (Normal)
	View (Normal)
	Delete (Normal)
	View (Highlighted)
	Delete (Highlighted)
	Previous Page (Normal)
	Next Page (Normal)
	Back (Normal)

Icon	Icon Meaning
	Rename (Normal)
	Edit (Normal)
	Confirm (Normal)
	Add Program (Normal)
	Delete Program (Normal)
	Settings (Normal)
	Power-on Preset (Normal)
	Export (Normal)
	Check (Normal)
	Log (Normal)

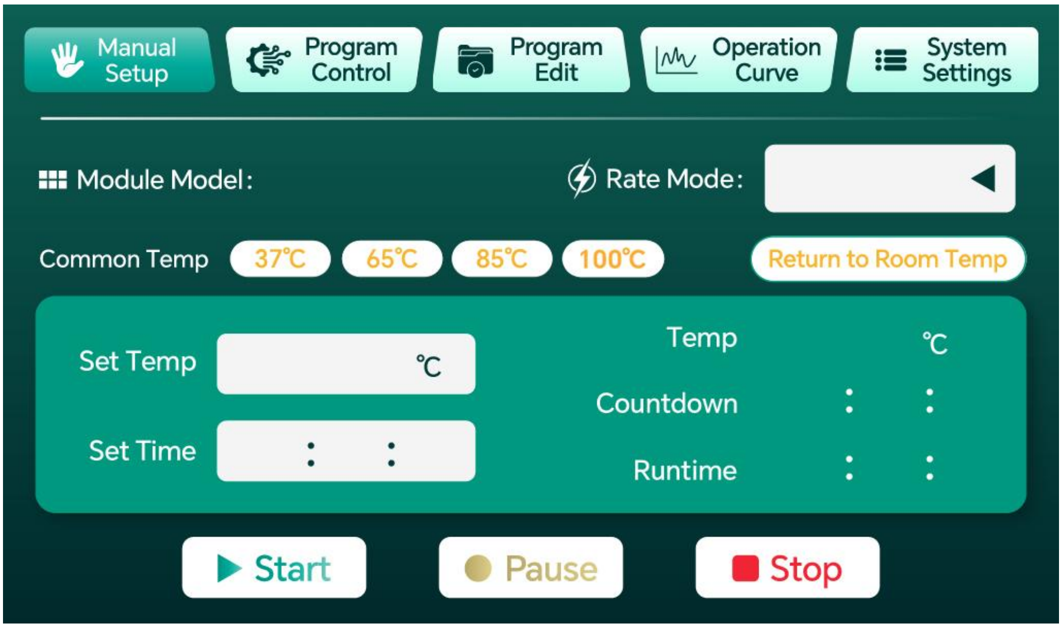
Icon	Icon Meaning
	Chinese/English (Normal)
	Heated Lid Mode (Normal)
	Alarm Information (Normal)
	Calibration (Normal)
	Settings (Highlighted)
	Power-on Preset (Highlighted)
	Export (Highlighted)
	Check (Highlighted)
	Log (Highlighted)

Icon	Icon Meaning
	Chinese/English (Highlighted)
	Heated Lid Mode (Highlighted)
	Alarm Information (Highlighted)
	Calibration (Highlighted)
	Cold Storage, Startup Preset, Hot Lid, Touch Beep Status ON (Highlighted)
	Cold Storage, Startup Preset, Hot Lid, Touch Beep Status OFF (Disabled)
	Log Details (Disabled)
	Log Details (Highlighted)
	Clear (Normal)
	Power-on Preset (Normal)
	Heated Lid Mode (Normal)

Icon	Icon Meaning
	Power-on Preset (Highlighted)
	Heated Lid Mode (Highlighted)

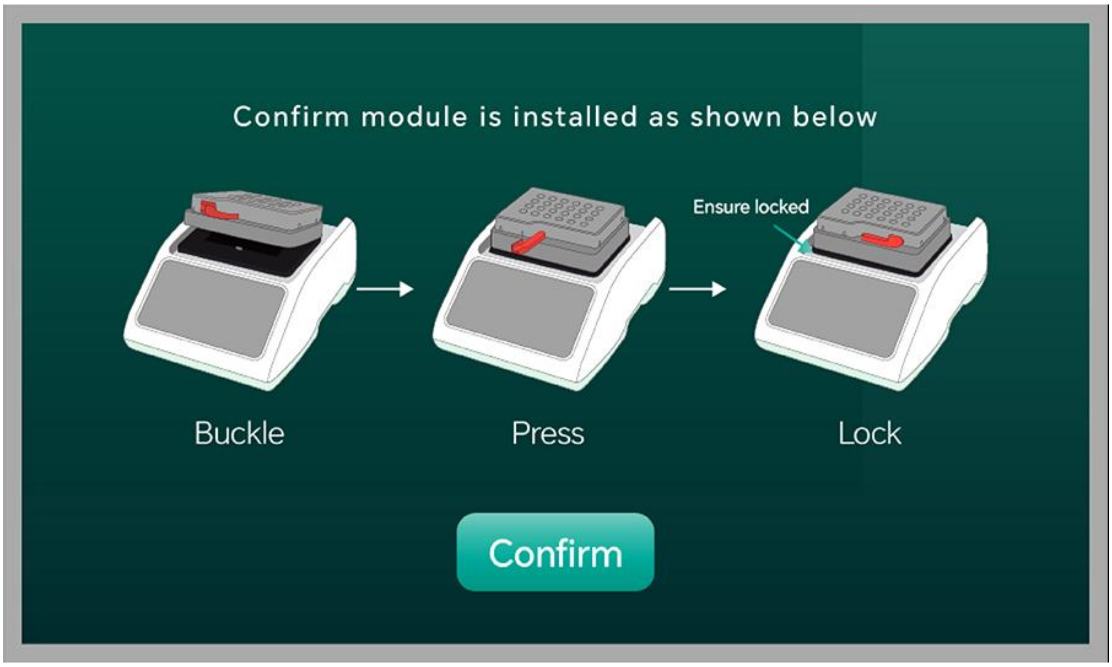
4.4.2 Startup Page

After the device is powered on, the screen first displays the boot screen (approximately 3 seconds), then automatically enters the power-on self-check process. The system will detect whether the temperature sensor is working properly within 10 seconds. After the self-check passes, it automatically enters the manual settings page.



After the self-check passes, the main manual settings page is displayed, and operation can begin.

Click the Run button and a prompt box will pop up, showing the module installation standard: 1 Snap - 2 Press - 3 Lock. After confirming that the installation is correct, click the Confirm button.

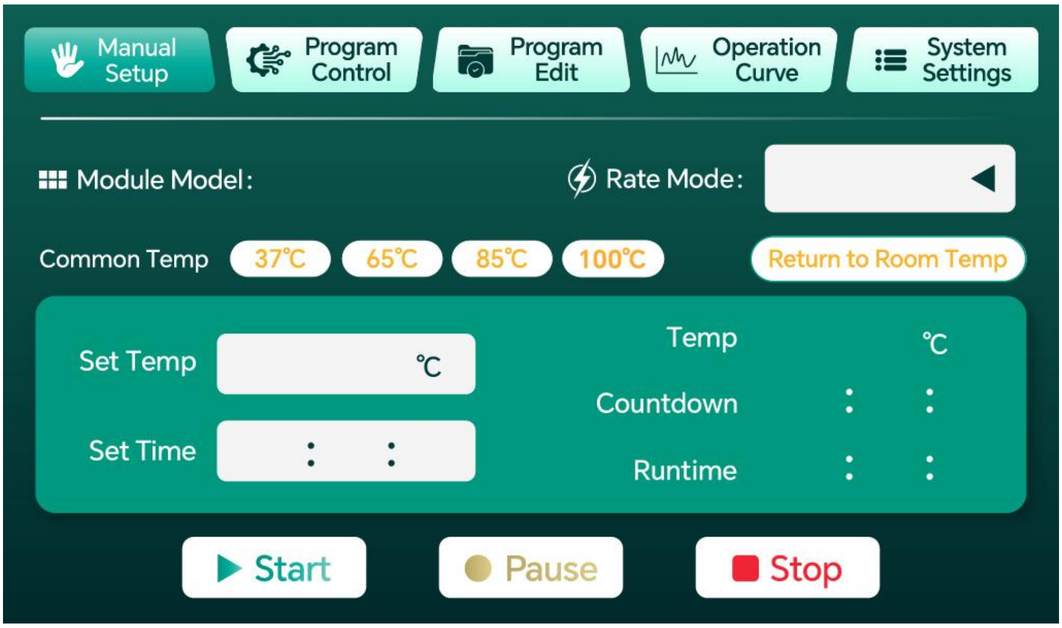


4.4.3 Manual Settings

The manual settings page is the main operating interface of the device. Users can complete temperature setting, time setting, rate mode selection, and device start/stop control on this page. It is divided into Page Overview, Temperature Setting, Time Setting, Rate Mode, Common Temperature Shortcuts, and Return to Room

Temperature.

4.4.3.1 Page Overview



● Module Model: Displays the currently detected module model (automatically identified via hardware ID pins, supports up to 16 module types).

● Rate Mode: Select from three heating/cooling rate levels: Basic Mode, High-Speed Mode, and Extreme Mode.

● Common Temperatures: Provides four shortcut buttons for commonly used temperatures (37°C, 65°C, 85°C, 100°C). Click to quickly set the target temperature.

● Return to Room Temperature: Returns the sample temperature to 23°C with one click. When the temperature reaches 23±1°C, it automatically stops, the button pops out, and the buzzer sounds once.

● Set Temperature: Click the input box to set the target temperature. Range: 23~100°C, accuracy: 0.1°C.

● Set Time: Set the hold time (H:MM:SS), maximum 99:59:59.

● Real-time Temperature: Displays the current real-time temperature of the sample area.

● Hold Countdown: Automatically starts countdown after the temperature reaches the set value.

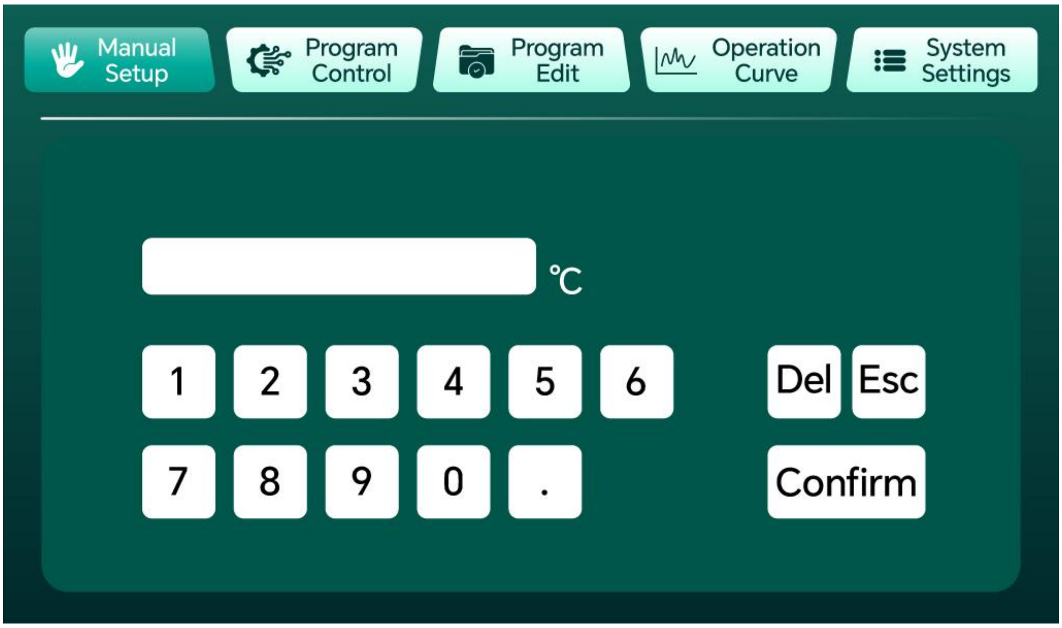
● Run Time: Displays the cumulative running time of the device.

● Run/Pause/Stop: Three control buttons for starting, pausing, and stopping the device.

Note: Both the temperature and time settings have a power-off memory function. After a power failure, the device will restore the previous settings upon restart.

4.4.3.2 Temperature Setting

Click the “Set Temperature” input box to bring up a numeric keypad. Enter the target temperature value and click “Confirm.” The temperature range is 23~100°C. An out-of-range value will trigger a prompt message.



4.4.3.3 Time Setting

Click the “Set Time” input box to bring up the time setting interface. Set hours, minutes, and seconds separately. Maximum settable time is 99:59:59. When the set time is 0, the device will run continuously (no countdown).

4.4.3.4 Rate Mode

Tap the “Rate Mode” dropdown to select from the following three heating/cooling rates:

Mode	Heating Rate	Cooling Rate	Recommended Use
Basic Mode	Approx. 2°C/min	Approx. 0.5°C/min	For experiments not requiring rapid temperature change
High-Speed Mode	Approx. 5°C/min	Approx. 1°C/min	For experiments requiring fast target temperature attainment
Extreme	Unlimited	Unlimited	Reach target temperature at

Mode	Heating Rate	Cooling Rate	Recommended Use
Mode			maximum speed

4.4.3.5 Common Temperature Shortcuts

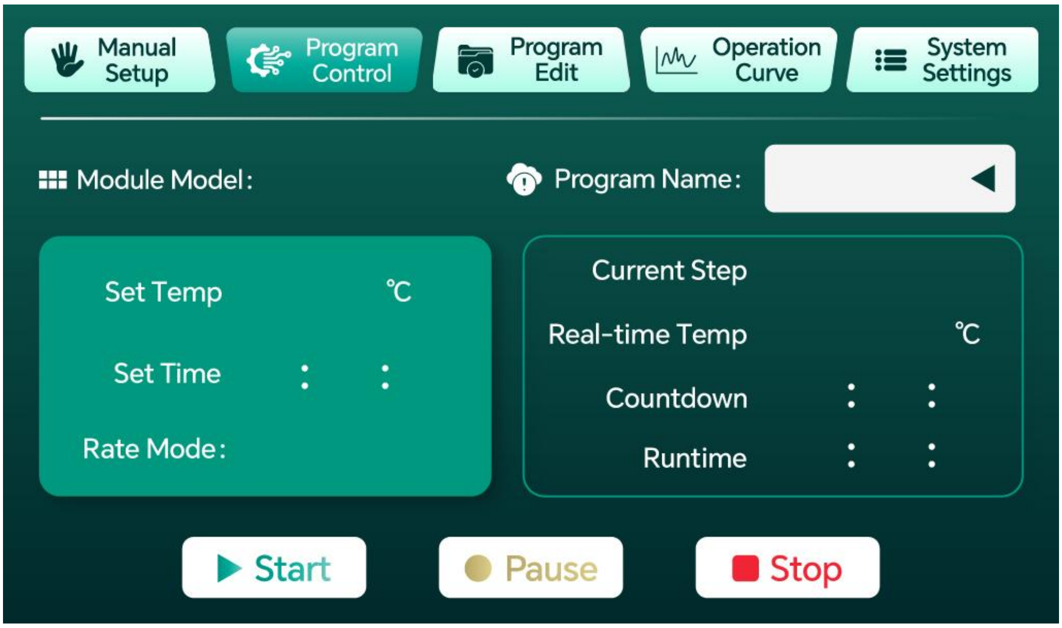
The manual settings page provides four shortcut buttons for commonly used temperatures: 37°C, 65°C, 85°C, and 100°C. Click the corresponding button, and the target temperature will be automatically set to that value without manual input. When the set temperature is not one of the common temperatures, the common temperature buttons pop out.

4.4.3.6 Return to Room Temperature

Click the “Return to Room Temperature” button, and the device will automatically regulate the temperature back to 23°C. When the temperature reaches 23±1°C, it automatically stops, the button pops out, and the buzzer sounds once. This function is suitable for safely removing samples after an experiment.

During the “Return to Room Temperature” process, click “Stop” to cancel this function. After stopping, the fan will continue to run for 10 seconds before turning off.

4.4.4 Program Control



● Set Temperature: Displays the target temperature of the current step (not directly editable; set from the Program Editing page).

● Set Time: Displays the hold time of the current step (H:MM:SS).

● Rate Mode: Displays the heating/cooling rate mode for the current step.

● Program Name: Dropdown menu in the upper-right corner to select the program to run (also selectable from the Program Editing page).

● Current Step: Displays the step number currently being executed.

● Real-time Temperature: Displays the current real-time temperature of the sample area.

● Hold Countdown: Countdown after reaching the temperature of the current step.

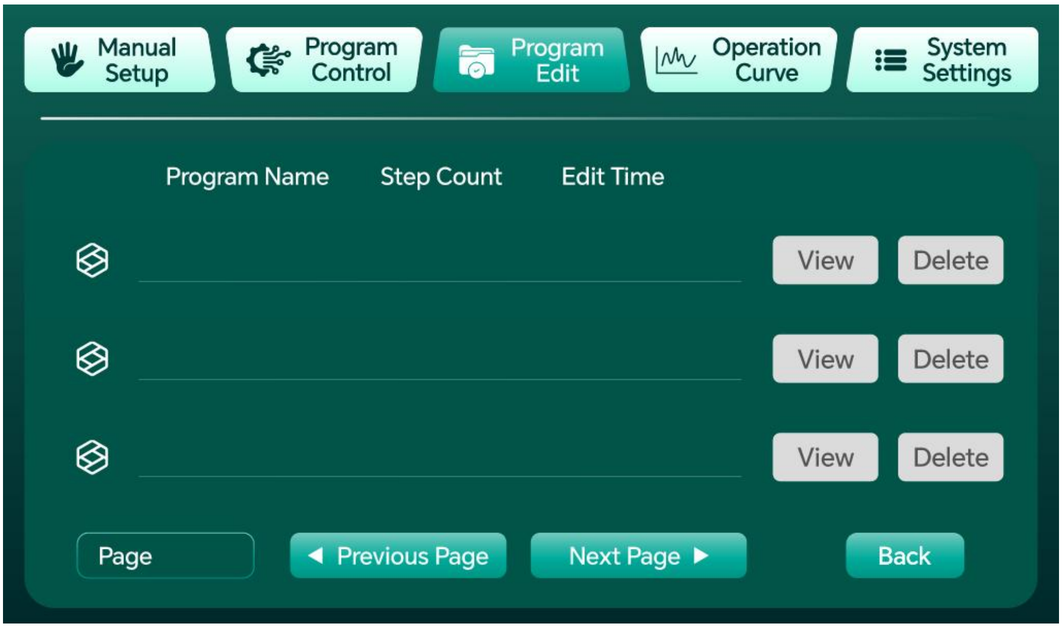
● Run Time: Displays the cumulative running time in program mode.

● Run/Pause/Stop: Controls the start and stop of the program.

Note: In program mode, when the current step countdown ends, it will automatically switch to the next step. After all steps are completed, the device will automatically stop and emit a buzzer alert.

4.4.5 Program Library

4.4.5.1 Program Library Page



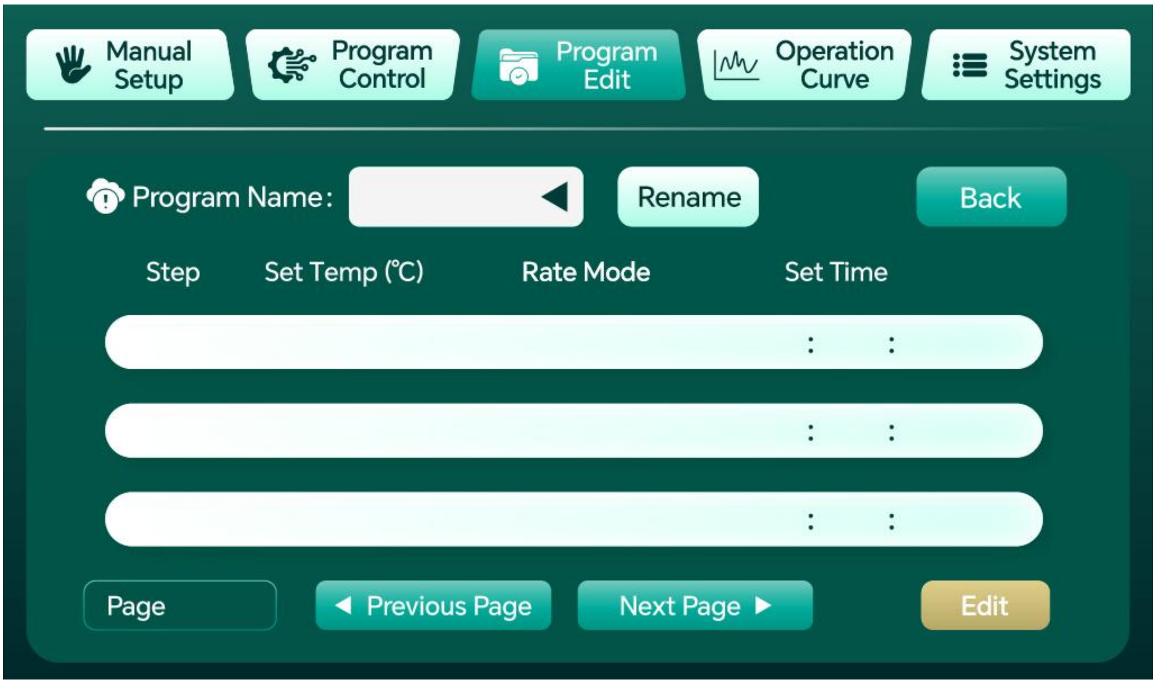
On the Program Library page, you can select a program to run and view detailed program steps. Maximum of 50 programs, each containing up to 10 steps. The set programs have a power-off memory function. After a power failure, the device will restore the previous settings upon restart.

● Click the “View” button to view the detailed content of the current program.

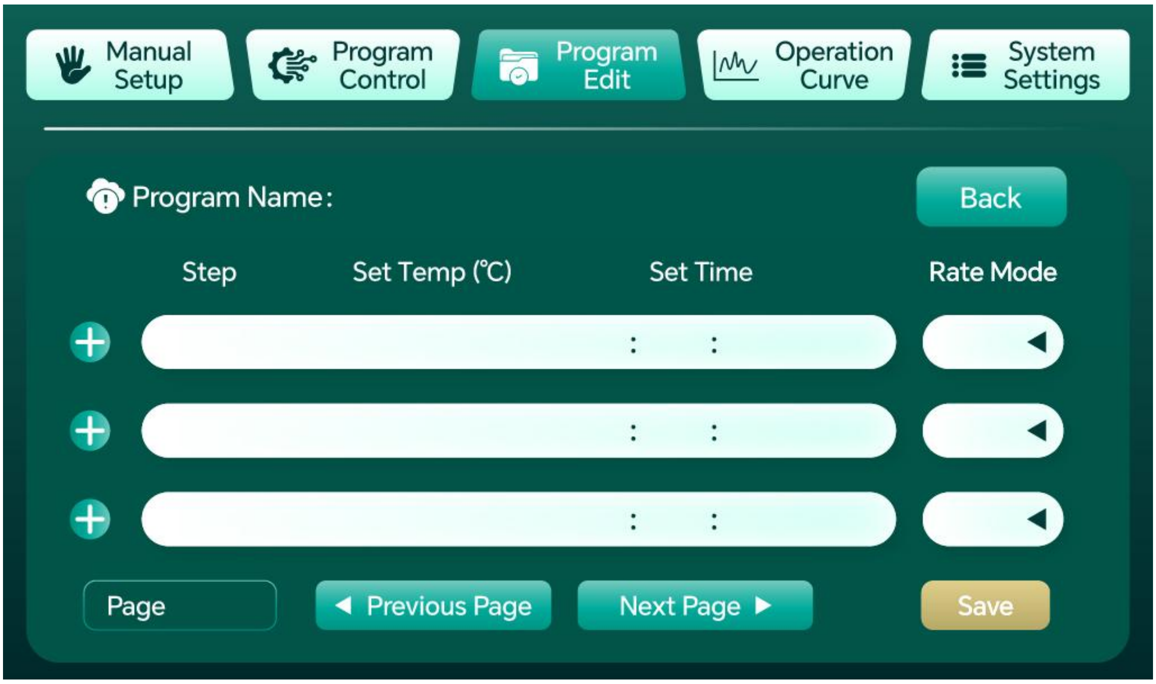
● Click the “Delete” button to delete the corresponding program.

4.4.5.2 Program Viewing

After clicking the “View” button, you enter the Program Viewing page, which displays all step details of the program (target temperature, hold time, rate mode). Click the “Program Edit” button to jump to the Program Editing page to edit the current program. The program name can be changed via the dropdown menu.



4.4.5.3 Program Editing Page



(1) The left side of the page has three (+) buttons. When a button is pressed, the

corresponding row text fields become editable. When the button is released, the corresponding row content is deleted and subsequent steps shift forward automatically. (10 steps max; must be added in sequence; the next button is disabled until the previous one is pressed.)

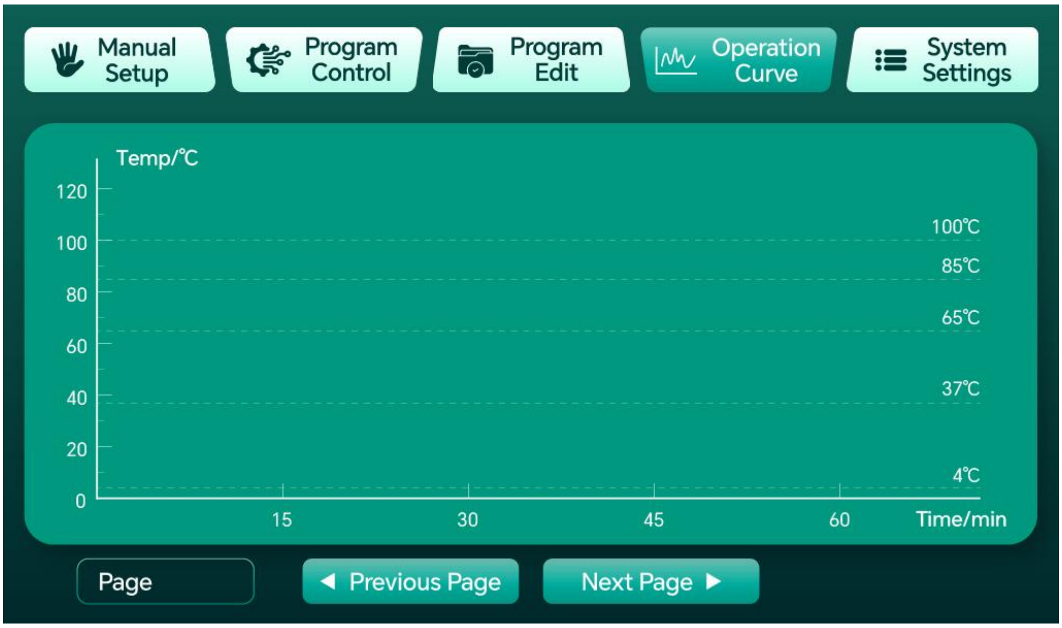
(2) Steps are paginated via “Previous Page” / “Next Page”. Each program can save a maximum of 10 steps.

(3) After editing, tap “Save” to save all step settings for the current program.

If the time or temperature is 0, saving is not allowed and a popup warning appears. A popup also confirms a successful save. Temperature must be within 23~100°C, and the set time cannot be 0; otherwise, a warning will pop up.

Note: Tap “Save” to return to the Program Library page.

4.4.6 Operation Curve



The running curve page is used to display temperature change trends in real time, helping users intuitively understand the temperature control process.

● The vertical axis is temperature (°C), and the horizontal axis is time (min).

● Common temperature reference lines are marked on the right side: 4°C, 37°C, 65°C, 85°C, 100°C.

● The curve can record up to 12,800 data points. Each page displays approximately 1 hour of data (720 points).

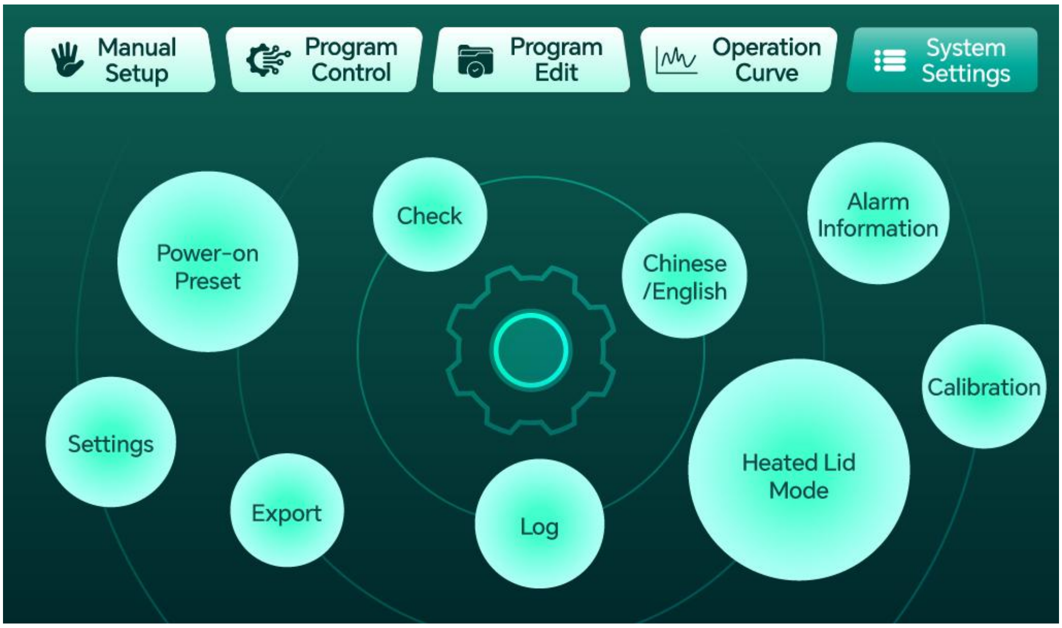
● Navigate through the Page X/Previous Page/Next Page buttons to view historical temperature trends. Supports quick jump navigation.

● The curve updates in real time during operation. Historical data can be reviewed after stopping.

Note: Page navigation (Page X/Previous Page/Next Page) is only available after the current page has finished rendering.

4.4.7 System Settings

4.4.7.1 Page Overview



(1) When clicking System Settings during manual settings or program operation, all icons except Settings and Calibration are disabled.

(2) After entering the Settings or Calibration menu from System Settings, the Manual Settings and Program Editing menus are grayed out.

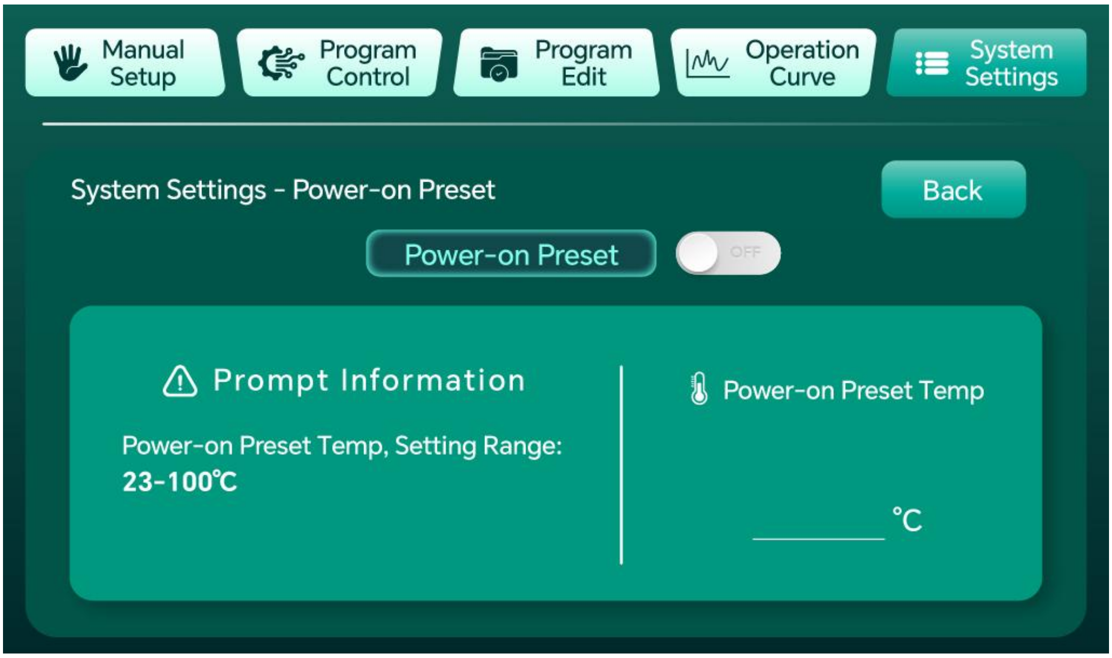
The System Settings page uses a circular icon layout and includes the following sub-function modules:

No.	Function Module	Description
1	Power-on Preset	Automatically pre-heats/pre-cools to the set temperature after power-on
2	Check	Re-execute the power-on self-check process
3	Chinese/English	Interface language switch (not yet enabled in current version)
4	Heated Lid Mode	Independent control of hot lid temperature

No.	Function Module	Description
5	Alarm Information	View historical alarm records
6	Calibration	Temperature compensation calibration
7	Settings	RTC time setting, restore default settings
8	Export	USB/SD card data export
9	Log	View running log records

Note: When entering the Calibration page from System Settings, a password is required.

4.4.7.2 Power-on Preset



Power-on preset refers to the function whereby the instrument automatically executes a pre-set constant temperature program after power-on, allowing the instrument to quickly reach a common standby temperature (such as 25°C or 37°C), reducing the user's waiting time before experiments.

● Power-on Preset Switch: ON/OFF toggle.

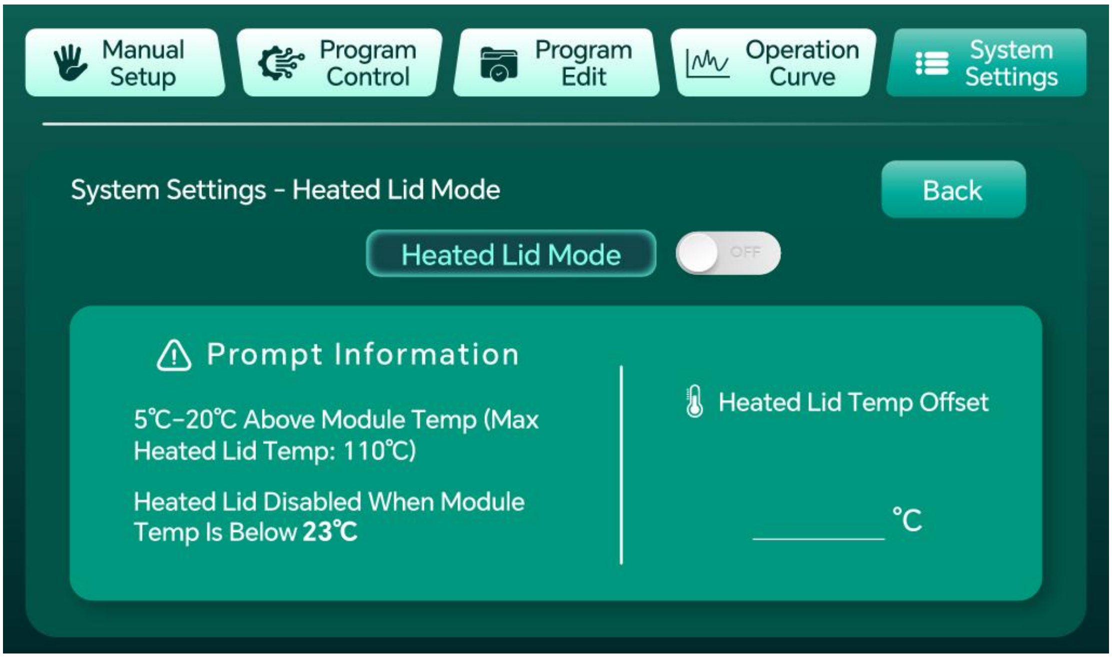
● Power-on Preset Temperature: Setting range 23~100°C. When enabled, the device automatically starts temperature control to reach the preset temperature upon power-up.

Operating Logic:

● Auto-start on Power-up: After the instrument is powered on and passes the self-check, if the power-on preset switch is detected as enabled, it will directly enter the preset constant temperature running state without user intervention.

● Priority and Interruption: This mode has lower priority than the user's main program. During power-on preset operation, once the user starts a new main program (e.g., Program A), the power-on preset will be immediately terminated, and the instrument will prioritize executing the user's specified main program.

4.4.7.3 Heated Lid Mode



The hot lid mode is used to independently control the temperature of the hot lid, preventing condensation on the inner wall of the sample lid during experiments such as PCR.

● Hot Lid Mode Switch: ON/OFF toggle.

● Hot Lid Temperature Differential: Set the temperature difference between the hot lid and the module temperature (5~20°C), not exceeding 110°C.

● Note: When the module temperature is below 23°C, the hot lid function is disabled.

Heated Lid Mode Operation Logic:

● Manual Mode: Turn on the hot lid mode switch and start the run program. The hot lid icon changes from white to yellow, indicating the hot lid function is active. Tap the Stop button to change the icon back from yellow to white.

● Program Control Mode: In Program Control Mode with the hot lid switch on, the device controls the hot lid according to the program target temperature: If the target temperature > 23°C, the hot lid activates and the icon changes from white to yellow; if the target temperature < 23°C, the hot lid does not operate and the icon stays white. Press Stop to change the yellow icon back to white.

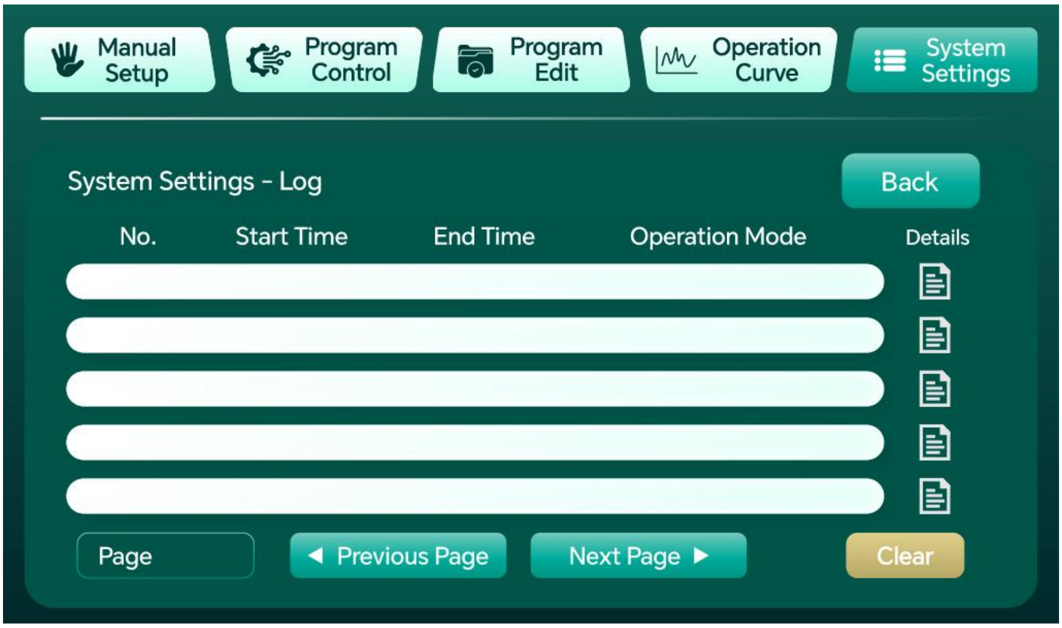
● If the hot lid is lifted during program operation, the system displays a popup: "Hot lid not detected. Continue running?"

Tap [Yes]: Disable hot lid function, hide hot lid status icon, program continues (running without hot lid mode);

Tap [No]: Disable hot lid function, hide hot lid status icon, terminate the current program.

Note: When hot lid mode is active, Target Temperature = Current Target Temperature - Hot Lid Temperature Differential × 5%.

4.4.7.4 Log Page



(1) Each row displays: No., Start Time (H:M:S), End Time (H:M:S), Run Mode (Manual Mode/Program Mode), and Details.

(2) Click the detail icon to enter the Log Detail page, showing detailed operation information for that log entry.

(3) Maximum of 100 log entries. Navigate via "Previous Page"/"Next Page" buttons.

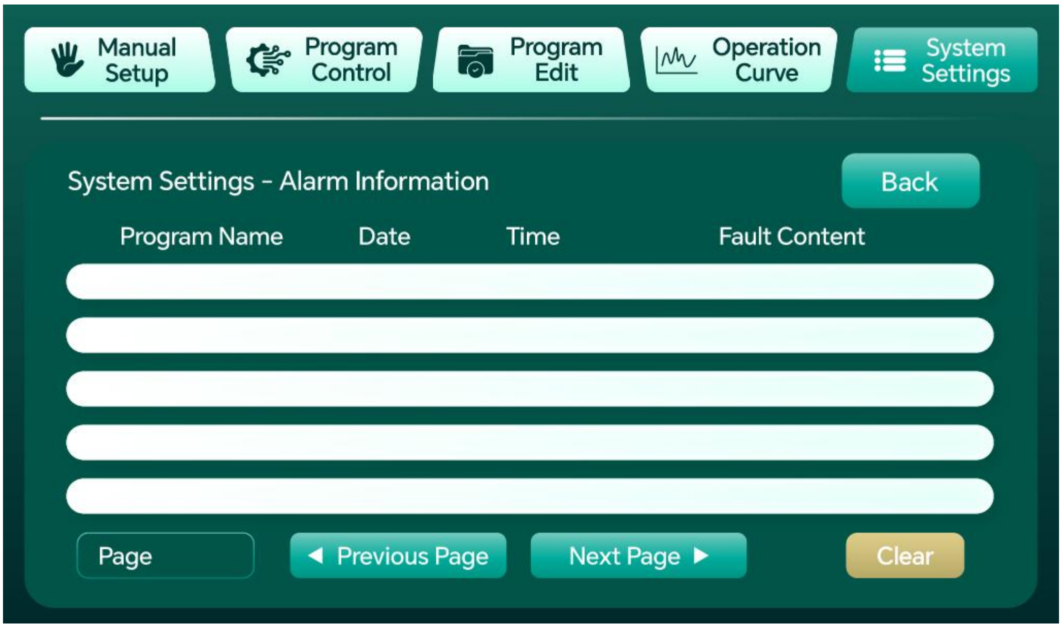
(4) Logs have a power-off memory function.

(5) Click the "Clear" button to clear all log entries. (Note: This process takes some time.)

Note: The Log Detail page is divided into Manual and Program tabs.

Important Note: After clicking the “Clear” button, log clearing will take some time. It is recommended to restart the instrument after the operation is complete.

4.4.7.5 Alarm Page



(1) Displays alarm information. Maximum of 100 entries. Navigate via “Previous Page”/“Next Page” buttons.

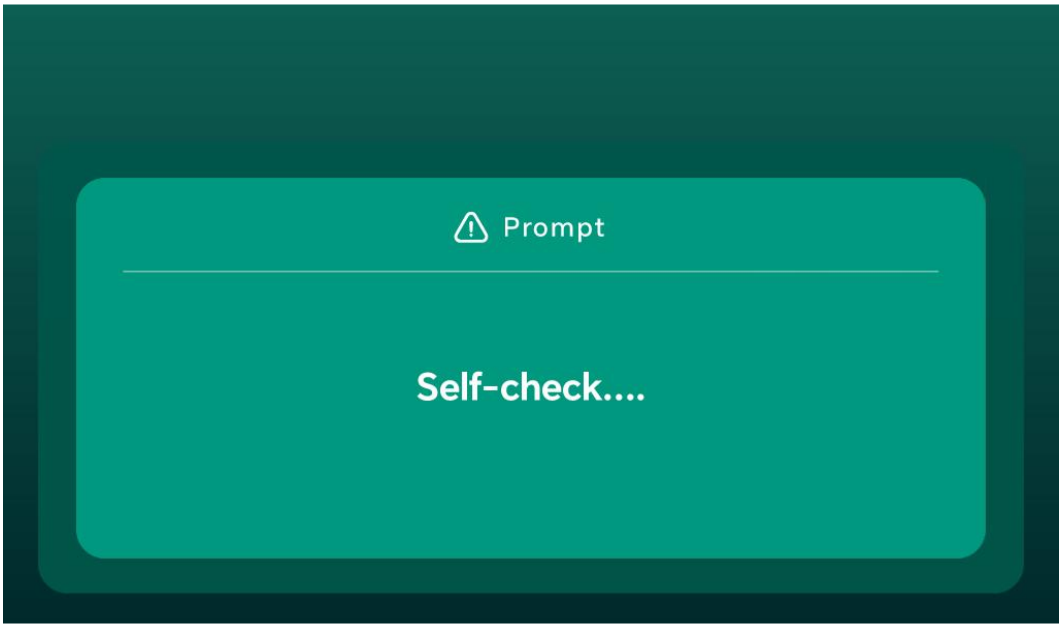
(2) Alarm information has a power-off memory function.

(3) Click the “Clear” button to clear all alarm entries. (Note: This process takes some time.)

Alarm types include:

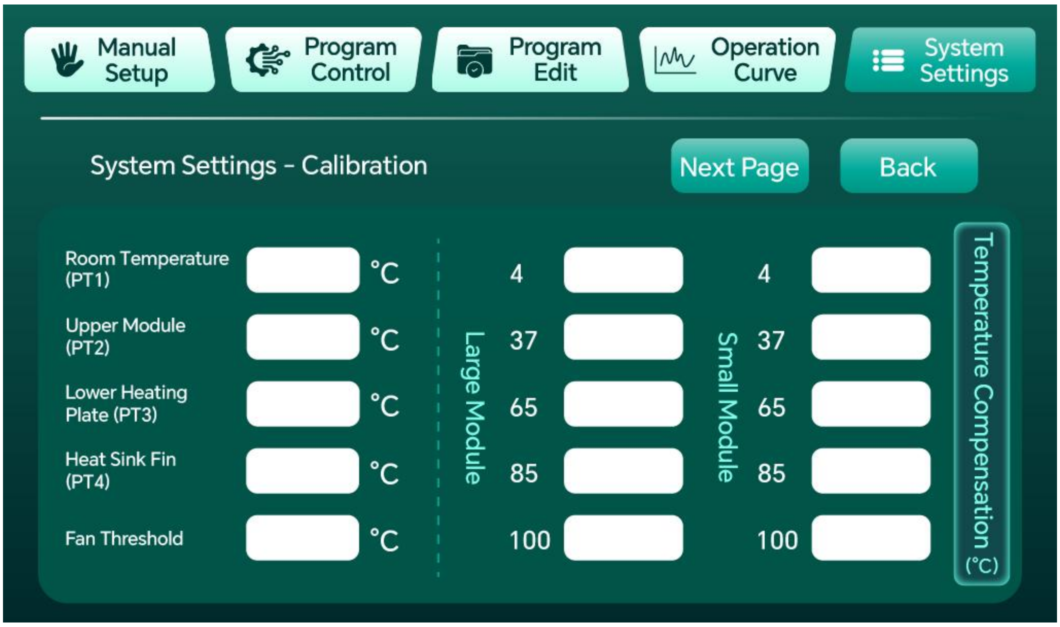
Alarm Type	Description
Temperature Anomaly	Temperature remains unchanged for an extended period during operation and has not reached the target temperature (if the temperature change is <0.05°C for 120 consecutive seconds, a temperature anomaly alarm is triggered)

4.4.7.6 Self-Check



Click the “Self-check” button, and the device will re-execute the power-on self-check process, checking whether the temperature sensor is working properly (whether valid temperature data is read within 10 seconds). After the self-check passes, it returns to the System Settings page. The fan runs at 50% speed during the self-check.

4.4.7.7 Calibration (Temperature Compensation)



The calibration function is used to compensate and calibrate temperature measurement values, improving temperature control accuracy. The system supports 4 groups of temperature compensation values, corresponding to 4 temperature zones:

Temperature Zone	Range
Low Temp Zone	4~15°C (including 4°C)
Room Temp Zone	15~55°C (including 15°C and 55°C)
Medium Temp Zone	55~85°C
High Temp Zone 1	85~100°C (including 85°C)
High Temp Zone 2	100°C

Users can adjust the compensation value for the corresponding zone based on the reading of a standard thermometer to make the displayed temperature more accurate.

· 5 mL×8, 15 mL×8, and 50 mL×4 are large modules and use large module temperature compensation.

· 0.6 mL×24, 1.5 mL×24, and 2.0 mL×24 well-plate modules are small modules and use small module temperature compensation.

4.4.7.8 Settings

The Settings page includes the following functions:

● RTC Time Setting: Set the real-time clock of the device (Year/Month/Day/Hour/Minute), providing accurate timestamps for log and alarm records.

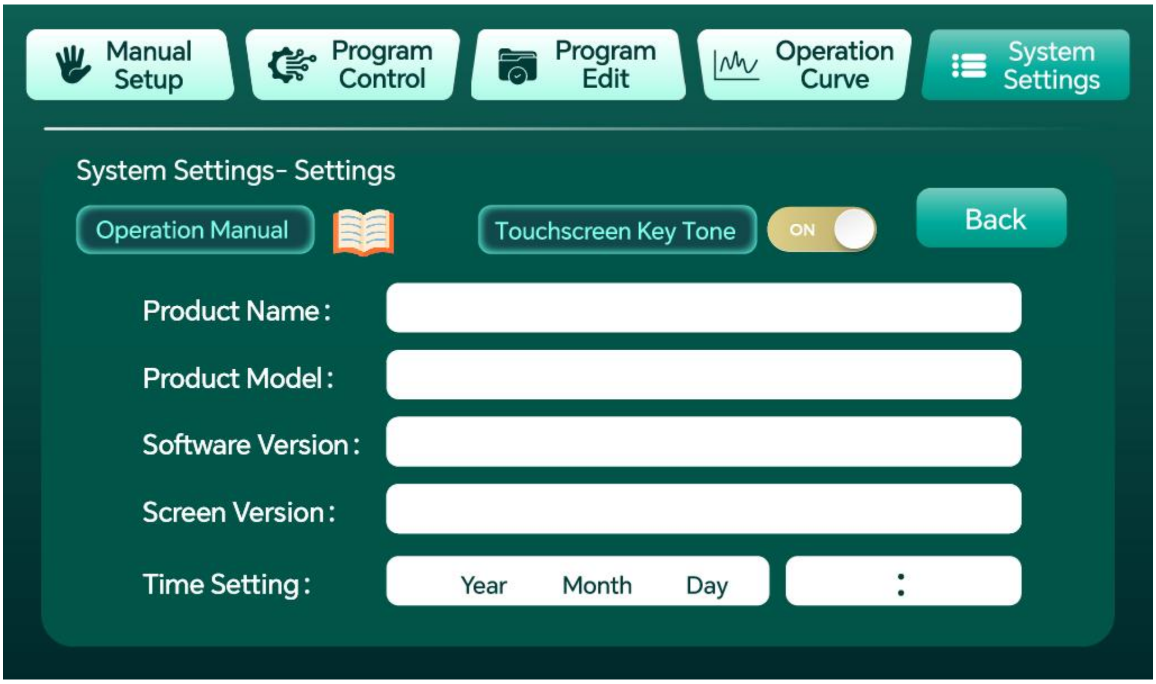
● Restore Default Settings: Restore all parameters to factory default values.

4.4.7.9 Data Export

● USB Export: Supports exporting running log data to a USB drive (Excel spreadsheet, CSV format) via the USB interface, for convenient data analysis on a computer.

● SD Card Auto-save: Supports periodic automatic saving of temperature data to the SD card, once every 5 minutes.

4.4.7.10 Product Information



Displays basic product information:

Item	Content
Product Name	B5 PRO Intelligent Metal Bath
Product Model	B5 PRO
Software Version	B5.PRO-V1.0.7
Screen Version	V1.0.5
Time Setting	Set the machine's year, month, day, hour, and minute

4.4.8 Popup Prompts

(1) Temperature Out of Range: A prompt will pop up when the set temperature exceeds the 23~100°C range.

(2) Module Not Detected: When no module is installed or the module is not properly installed, the module model displays “Module Not Detected,” and the Run/Pause/Stop buttons are all disabled (both manual and program modes).

(3) Hot Lid Not Detected: If the hot lid is lifted during program operation, a popup will appear: “Hot lid not detected. Continue running?”

(4) Program Save Failed: If the time or temperature is 0, saving is not allowed and a popup warning appears. Temperature must be within 23~100°C, and the set time

cannot be 0; otherwise, a warning will pop up.

(5) Program Saved: A popup also confirms a successful save.

4.5 Function Description

4.5.1 Two Operating Modes

(1) Manual Mode: Operation method - Set the target temperature and hold time on the Manual Settings page, then click the “Run” button to start. Stop method - Click the “Stop” button, or the device automatically stops when the hold time expires.

(2) Program Mode: Operation method - Select or edit a program on the Program Editing page, set the target temperature, hold time, and rate mode for each step, then click “Run.” Stop method - Click the “Stop” button, or the device automatically stops when all steps are completed.

4.5.2 Manual Mode

4.5.2.1 Prerequisites

① Operation Mode: Manual Settings page
② Set the target temperature: 23~100°C
③ Set the hold time (hold timing starts after the set temperature is reached; the device automatically stops when the hold time ends)
④ Select the rate mode (Basic/High/Extreme)

4.5.2.2 Operation Procedure

① Click “Run” - the device starts heating/cooling to the target temperature. The Run button lights up, and the Stop button turns off. (The buzzer sounds once.)

② When the temperature reaches the set value, the hold countdown automatically starts.

③ When the hold countdown ends, the buzzer beeps three times, and the device automatically stops.

④ During operation, you can click the “Pause” button at any time to pause, and click “Run” again to resume.

⑤ During operation, you can click the “Stop” button at any time to stop.

Note: 1. If a temperature anomaly occurs during operation, the device will automatically stop and display alarm information on the alarm page, and records the alarm information.

2. Power-off Resume Function: If an unexpected power failure occurs during

operation, the device will automatically resume operation from the breakpoint when power is restored.

3. During operation, the temperature and time input boxes cannot be edited, and the rate mode cannot be modified.

4.5.3 Program Mode

4.5.3.1 Prerequisites

① Operation Mode: Program Control page
② Select the program to run on the Program Editing page
③ Confirm that the step parameters (target temperature, hold time, rate mode) are correctly set

4.5.3.2 Operation Procedure

① Click “Run” - the device starts executing the first step, heating/cooling to the target temperature of that step. (The buzzer sounds once.)

② When the temperature of the current step is reached, the hold countdown starts. When the countdown ends, the device automatically switches to the next step, and the buzzer beeps once to indicate step switching.

③ The current step number, target temperature, and hold countdown are updated in real time.

④ After all steps are completed, the buzzer beeps three times, and the device automatically stops.

Note: 1. During operation, click the “Stop” button at any time to stop.

2. When restarted after stopping, the current step, target temperature, and hold time will be restored to those of Step 1 of the program.

3. Power-off Resume Function: If an unexpected power failure occurs during operation, the device will automatically resume from the breakpoint (including the current step and remaining time) when power is restored.

4. The program name cannot be changed during program operation.

5. The lower heating plate temperature must not exceed 108°C (physical limit; output may stop if exceeded).

4.6 Operation Instructions

4.6.1 Module and Lid Installation

Module replacement is simple and requires no tools. After proper installation, the system automatically identifies and displays the module model via hardware ID pins; if the module model displays “Module Not Detected,” please reinstall the module.

When using the optional heated lid, simply place it on top of the corresponding module (1.5/2mL); note that the hot lid function is disabled when the module temperature is below 20°C.

Note: Module installation steps: 1 Snap — 2 Press — 3 Lock

4.6.2 Daily Power-On/Off

1. Before turning off the instrument, stop the device first and wait for the temperature to return to room temperature before switching off the power.

2. Do not restart the instrument immediately after turning off the power. Improper operation will affect the service life of the instrument.

4.6.3 Scan QR Code to View Instructions

(Scan with WeChat or mobile browser to view instrument operation instructions)

5. Maintenance

5.1 Routine Maintenance

5.1.1 Surface Cleaning

5.1.1.1 Do not attempt to clean the instrument while the power cord is plugged in or the power switch is in the ON position;

5.1.1.2 Immediately after use, wipe the instrument surface with a soft cloth or alcohol swab to prevent sample residue (e.g., chemical reagents, biological fluids). Never wipe the module directly with a wet cloth at high temperatures, to prevent module damage caused by thermal stress;

5.1.1.3 Stains can be cleaned with a neutral detergent; avoid corrosive solvents (e.g., strong acids, strong bases);

5.1.2 Module and Heated Lid Maintenance

5.1.2.1 Periodically check whether the module is properly installed. If the module model displays “Module Not Detected,” please reinstall it;

5.1.2.2 Module replacement is simple and requires no tools. Do not strike the module or heated lid with sharp objects, to prevent cracks in the module and main unit during use caused by scratches or external damage;

5.1.2.3 Wait for the heated lid to cool down before cleaning and storing it.

5.1.3 Power Cord Inspection

5.1.3.1 Check the power cord for damage or poor contact to avoid short circuit risks.

5.2 Periodic Maintenance

5.2.1 Periodic Self-Check

5.2.1.1 Periodically re-execute the power-on self-check process via "System Settings → Self-check" to verify whether the temperature sensor is working properly (whether valid temperature data is read within 10 seconds);

5.2.1.2 After the self-check passes, it returns to the System Settings page. The fan runs at 50% speed during the self-check.

5.2.2 Internal Dust Removal

5.2.2.1 Periodically inspect the instrument's cooling vents and remove dust and debris to prevent effects on heat dissipation or circuit performance;

5.2.2.2 Caution: Power must be disconnected first, and the instrument must be allowed to return to room temperature before operation. It is recommended that this be performed by professional personnel.

5.2.3 Temperature Calibration

5.2.3.1 It is recommended to periodically calibrate the instrument temperature using a standard thermometer (System Settings → Calibration), adjusting the compensation value of the corresponding temperature zone to make the displayed temperature more accurate.

5.2.4 Disassembly and Transport

Do not strike the module or heated lid with sharp objects. Prevent impacts during transport, disassembly, and assembly to avoid cracks in the module and main unit during use caused by scratches or external damage.

6. Troubleshooting

6.1 Temperature Anomaly Alarm

6.1.1 Possible Causes

The temperature remains unchanged for an extended period during operation and has not reached the target temperature (if the temperature change is <0.05°C for 120 consecutive seconds, a temperature anomaly alarm is triggered).

6.1.2 Troubleshooting Steps

① The device will automatically stop and display alarm information on the alarm page, and records the alarm information; ② Check whether the module is properly installed and whether the sample load is normal; ③ Restart the device. If the alarm persists, stop using the instrument and contact the manufacturer or authorized repair personnel.

6.2 Module Not Detected

6.2.1 Possible Causes

The module is not installed or is not properly installed.

6.2.2 Troubleshooting Steps

① Check whether the module model on the Manual Settings page displays “Module Not Detected”; at this time the Run/Pause/Stop buttons are all disabled; ② Reinstall the module and ensure it is properly seated (module replacement requires no tools); ③ After proper installation, the module model display is restored, and the Run/Pause/Stop buttons become available again.

6.3 Hot Lid Anomaly

6.3.1 Possible Causes

The hot lid was lifted during program operation, or the module temperature is below 20°C (hot lid function disabled).

6.3.2 Troubleshooting Steps

① When the popup “Hot lid not detected. Continue running?” appears, tap [Yes] to disable the hot lid function and continue running (no hot lid mode), or tap [No] to terminate the current program; ② Confirm that the hot lid is correctly placed on top of the corresponding module (1.5/2mL); ③ When the module temperature is below 23°C, wait for the temperature to rise before using the hot lid function.

6.4 No Heating / Output Stopped

6.4.1 Possible Causes

The lower heating plate temperature exceeds 108°C (physical limit; output may stop if exceeded).

6.4.2 Troubleshooting Steps

① Click the “Stop” button and wait for the instrument temperature to drop; ② Check whether the set temperature matches the actual load; ③ Reset the temperature and restart. If the instrument still does not work properly, contact the manufacturer or authorized repair personnel.

7. Usage Precautions

7.1 Operating Restrictions

7.1.1 During operation, the temperature and time input boxes cannot be edited, and the rate mode cannot be modified;

7.1.2 The program name cannot be changed during program operation;

7.1.3 The lower heating plate temperature must not exceed 108°C (physical limit; output may stop if exceeded);

7.1.4 When the set time is 0, the device will run continuously (no countdown).

7.2 Module Usage

7.2.1 When no module is installed, the module model displays "Module Not Detected," and Run/Pause/Stop are all grayed out (both manual and program modes);

7.2.2 When the module temperature is below 23°C, the hot lid function is disabled;

7.2.3 When hot lid mode is active, Target Temperature = Current Target Temperature - Hot Lid Temperature Differential × 5%.

7.3 Power-Off Resume

7.3.1 If an unexpected power failure occurs during operation, the device will automatically resume operation from the breakpoint when power is restored (in program mode, including the current step and remaining time);

7.3.2 Do not restart the instrument immediately after turning off the power. Improper operation will affect the service life of the instrument.

7.4 Calibration Recommendations

7.4.1 It is recommended to periodically verify the temperature using a standard thermometer (System Settings → Calibration), adjusting the compensation value of the corresponding temperature zone to ensure the displayed temperature matches the actual temperature (especially important for experiments with high precision requirements).

7.5 Professional Maintenance Recommendations

7.5.1 If the instrument cannot be repaired through simple troubleshooting, contact the manufacturer or an authorized service center; avoid self-disassembly to prevent secondary damage;

7.5.2 Retain the purchase receipt and warranty card; free periodic maintenance services are provided.

8. Module List

No.	Module Name	Compatible Containers	Specification	Remarks
1	Centrifuge tube module	0.6mL centrifuge tubes	0.6mL×24	Small module temperature compensation
2	Centrifuge tube module	1.5mL centrifuge tubes	1.5mL×24	Small module temperature compensation
3	Centrifuge tube module	2.0mL centrifuge tubes	2.0mL×24	Small module temperature compensation
4	Centrifuge tube module	5mL centrifuge tubes	5mL×8	Large module temperature compensation
5	Centrifuge tube module	15mL centrifuge tubes	15mL×8	Large module temperature compensation
6	Centrifuge tube module	50mL centrifuge tubes	50mL×4	Large module temperature compensation
7	Multi-well plate module	Cell culture plates, ELISA plates, full-skirted microplates	--	Tool-free module replacement
8	Standard module lid	--	--	Standard (precision temperature control)
9	Heated lid	1.5/2mL modules	--	Optional (prevents

No.	Module Name	Compatible Containers	Specification	Remarks
				low-temperature condensation)

Intelligent Metal Bath and Module Specifications:

Item No.	Product Name	Specification
QT-TMC-1290	B5 Pro Intelligent Metal Bath	1 unit/carton
QT-TMC-1291	B5 Max Intelligent Metal Bath	1 unit/carton
QT-B1	Module B1 (0.65mL centrifuge tubes ×24)	1 piece/box
QT-B2	Module B2 (1.5mL centrifuge tubes ×24)	1 piece/box
QT-B3	Module B3 (2mL centrifuge tubes ×24)	1 piece/box
QT-B4	Module B4 (5mL centrifuge tubes ×8)	1 piece/box
QT-B5	Module B5 (15mL centrifuge tubes ×8)	1 piece/box
QT-B6	Module B6 (50mL centrifuge tubes ×4)	1 piece/box
QT-B7	Module B7 (Flat plate type)	1 piece/box

Temperature control accuracy up to 0.2°C, heating/cooling efficiency improved by 50%+, and module replacement in just 5 seconds.

Let researchers focus on the experiment itself, not on the equipment.

9. After-Sales Service

9.1 Warranty Service

9.1.1 3-Year Whole-Unit Warranty: Free repair or replacement of core components such as the heating/cooling module and mainboard (for non-human-caused damage).

9.1.2 1-Year Screen/Touch Module Warranty: Free replacement for touchscreen unresponsiveness or display issues caused by normal use.

9.2 Repair Service

9.2.1 Rapid Response: Technical response to after-sales issues within 24 hours on working days, with a solution provided within 48 hours.

9.2.2 Nationwide Warranty: Supports mail-in repair or on-site service (Shanghai area supports on-site service within 2 working days).

9.2.3 Backup Unit Support: If the repair cycle exceeds 5 working days, a free backup unit can be requested.

9.3 Parts Replacement

9.3.1 Consumables Discount: During the warranty period, optional parts such as module lids and heated lids are eligible for 60% off replacement.

9.3.2 Lifetime Maintenance: After the warranty expires, only cost-price fees are charged, with official OEM parts provided.

9.4 Technical Support

9.4.1 Professional Consultation: Free professional technical support for experiment method optimization, troubleshooting, and more.

9.4.2 Remote Diagnostics: Rapid problem-solving guidance via app or video call.

9.5 Return and Exchange Policy

7-Day Quality Issue Replacement: Direct replacement with a new unit for non-human-caused performance failures within 7 days of receipt.

9.6 Value-Added Services

Free Calibration: One official performance calibration per year (requires return shipment or on-site appointment).

Device Warranty Card

Product Information

Product Name: Intelligent Metal Bath

Model: ____________

Serial No.: ____________

Purchase Date: ____________

Warranty Policy

Warranty Period: Free warranty service for 48 months from the date of purchase.

Warranty Coverage: Performance failures caused by material or manufacturing defects (e.g., heating/cooling abnormality, circuit board failure, sensor malfunction, etc.). Free repair or replacement of damaged parts during the warranty period (for non-human-caused damage).

Warranty Proof: This warranty card and a valid proof of purchase (invoice/receipt) are required.

Disclaimer (Not Covered Under Warranty)

The following situations require paid repair:

Human-caused damage (e.g., dropping, liquid infiltration, improper operation, etc.).

Use or maintenance not in accordance with the manual requirements, or unauthorized modification or disassembly of the product.

Damage caused by natural disasters or force majeure (e.g., fire, flood, lightning strike, etc.).

Failures caused by unauthorized third-party repairs.

Normal wear and tear or consumables (e.g., module lids, heated lids, and other wear parts).

Warranty Service Process

Fault Reporting:

Contact the customer service hotline/email, describe the fault symptoms, and provide the product serial number and proof of purchase.

After customer service confirmation, you will be informed of the repair drop-off or mail-in procedure.

Repair Processing:

After receiving the product, an engineer will inspect it and provide feedback within 5-7 working days.

Faults within warranty coverage are repaired free of charge; for non-warranty issues, repair proceeds after the user confirms the cost.

Product Return: After repair is completed, the product will be shipped back to the user's designated address (free shipping during the warranty period).

Contact Information

Service Hotline: 400-025-2120

Email: quanlab@lab-direct.com

Official Website: www.quan-lab.com

Repair Center Address: Building 21, Baijiatong, No. 388 Shengrong Road, Pudong New District, Shanghai

Precautions

Please keep this warranty card and proof of purchase properly; they cannot be reissued if lost.

Friendly Reminder: Regular cleaning and maintenance can extend the life of the device. Please refer to the manual for instructions.